

Zainab Habib Almahal

Junior Data Scientist | ML Engineer

Eastern Province, Saudi Arabia | [LinkedIn](#) | [GitHub](#)

SUMMARY

Computer Science graduate from IAU with AWS certifications in Machine Learning Engineering and AI, and hands-on experience building data warehouses, statistical models, and LLM-powered analytics tools. Comfortable across the full data science stack — SQL, Python (pandas, statsmodels, scikit-learn), BI dashboards, and applied NLP/LLM function calling — with a full-stack engineering background (React, ASP.NET Core, Laravel) for shipping production-ready data products. Bilingual in English and Arabic, with a track record of building bilingual EN/AR data products for the Saudi market.

SKILLS

Data Science & ML: Python (pandas, NumPy, statsmodels, scikit-learn, XGBoost), SQL (PostgreSQL, MySQL), statistical analysis & hypothesis testing, feature engineering, predictive modeling, AWS SageMaker

LLM / NLP: Google Gemini API, function calling, retrieval-augmented generation (RAG), embeddings, prompt engineering, OpenAI API

Data & Cloud Tools: Metabase, Docker, AWS (S3, Lambda, API Gateway, CloudWatch), Git

Software Development: React, TypeScript, ASP.NET Core (C#), Laravel (PHP), HTML/CSS, RESTful APIs, Java, C++

Languages: Arabic (Native), English (Fluent — IELTS 7.5 Academic)

WORK EXPERIENCE

Full-Stack Developer, Alelm Alsatea – Technology Solutions

08/2026 – Present

Dammam

- Develop and maintain web and mobile applications to meet business and user requirements
- Maintain, update, and optimize existing applications, troubleshooting and debugging to improve reliability and performance
- Design, build, and test new applications and features against project requirements

Software Developer – Intern, Nesma & Partners

06/2025 – 08/2025

Al Khobar

- Built responsive UIs using React, TypeScript, and Tailwind CSS
- Developed RESTful APIs with ASP.NET Core (C#)
- Designed and queried MySQL databases

PROJECTS

Credit Portfolio Warehouse – Financial Data Warehouse & Bilingual GenAI Query Layer

08/2026

Independent project simulating an end-to-end Saudi SME credit portfolio

- Built a Dockerized PostgreSQL data warehouse for a simulated Saudi SME credit portfolio (2,000 companies, 3,500 loans, SAR 2.18B outstanding), with SQL-based segmentation surfaced through a Metabase BI dashboard
- Applied statistical methods (logistic regression, Wilson confidence intervals, power analysis) in Python to validate portfolio risk drivers, including identifying and correcting a confounding variable
- Built a bilingual (EN/AR) GenAI natural-language query layer using Gemini function calling and embedding-based RAG, letting non-technical users query the warehouse in either language without writing SQL

BizStart Simulator – Bilingual AI Financial Risk & Runway Modeling Tool

07/2026

Kanz AI Training Hackathon (Guinness World Records event, 14,075 live participants)

- Built a bilingual (EN/AR) financial modeling tool for early-stage founders using React, TypeScript, and Recharts, generating interactive revenue/cost projections from a simple business idea input
- Designed an AI-driven risk-check feature flagging runway, margin, and marketing-spend risk against real failure-rate benchmarks; selected as an AI Showcase Featured Project at the hackathon

AWS ML Pipeline – Titanic Survival Prediction

06/2026

- Built an end-to-end ML pipeline on AWS (SageMaker, S3, Lambda, API Gateway) and deployed an XGBoost model as a real-time REST API with CloudWatch logging and monitoring

StrokeAlert – Stroke Detection ML Pipeline

07/2025

Graduation Project, Imam Abdulrahman Bin Faisal University (IAU)

- Developed an end-to-end ML pipeline for stroke detection using real CT scan data — preprocessing, feature engineering, model training, and evaluation

EDUCATION

Bachelor of Computer Science, Imam Abdulrahman Bin Faisal University (IAU)

08/2020 – 08/2025

Al Khobar | Second Honors | IELTS: 7.5 (Academic)

PROFESSIONAL CERTIFICATIONS

AWS Certified Machine Learning Engineer – Associate (06/2026 – 06/2029) | AWS Certified AI Practitioner (03/2026 – 03/2029)

COURSES / TRAINING

Introduction to Artificial Intelligence, KAUST Academy (01/2026) | Mathematics for Machine Learning and Data Science, DeepLearning.AI (12/2025) | Meta Front-End Developer, Meta (03/2025)